

Digital strategies for the social sciences – 10

Non supervised Machine Learning

Introduction

What is machine learning?

Machine learning in a non-supervised context

2 non-supervised algorithm in detail

Introduction

Questions?

MCA: another example

Questions about the final

WHAT IS MACHINE LEARNING?

A buzzword

A (poorly defined) class of statistical techniques

- Statistical learning, artificial intelligence, statistical learning, pattern recognition, sometimes data mining

WHAT IS MACHINE LEARNING?

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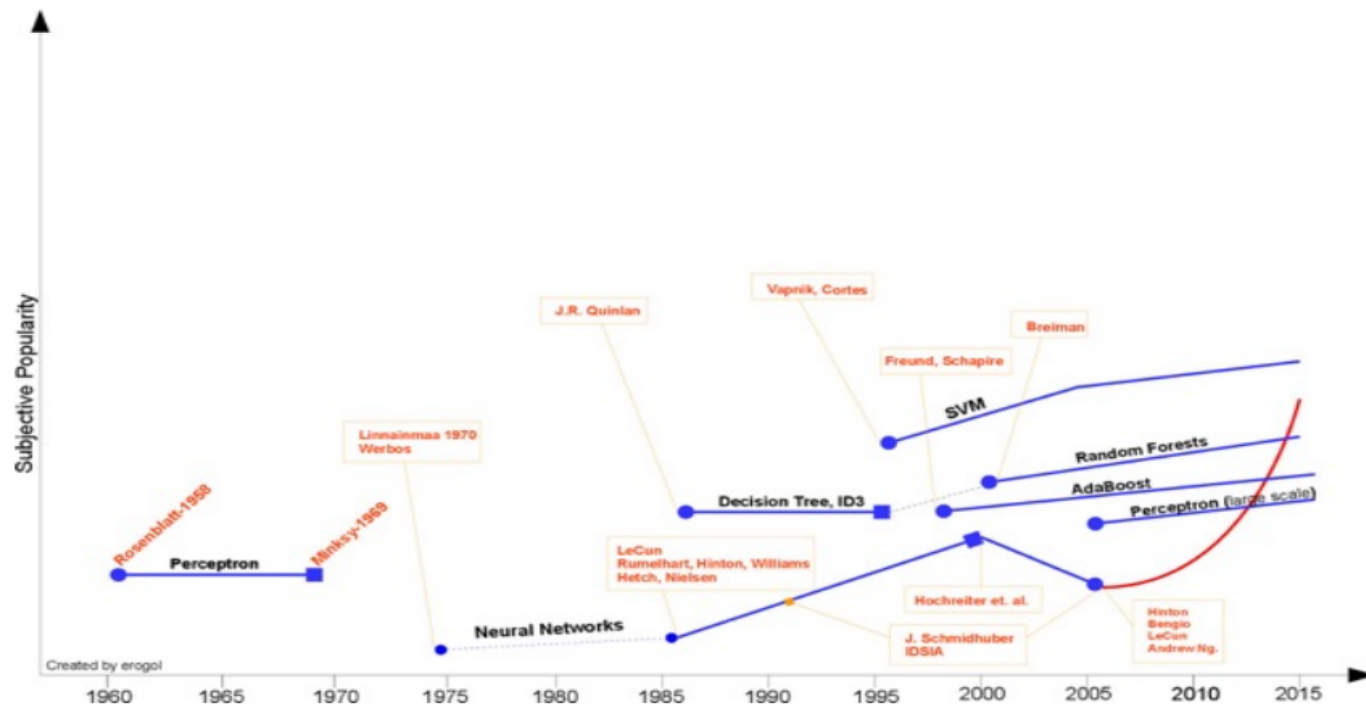
- Statistical learning, Artificial intelligence, Statistical learning, pattern recognition, sometimes data mining
- Varied definitions
 - “A set of methods that can automatically detect patterns in data” (Murphy, 2012)
 - “Machine learning is programming computers to optimize a performance criterion using example data or past experience” (Alpaydin, 2009)
 - “The field of study that gives computers the ability to learn without being explicitly programmed” (Samuel, 1952)

WHAT IS MACHINE LEARNING?

History

Is about to be written

- Starts in the 1950s
- A series of booms and busts



WHAT IS MACHINE LEARNING?

History

- Is partly technical
 - Role of the size of the data sets, of the computer power
- Is also partly commercial

WHAT IS MACHINE LEARNING?

What does machine learning do?

A lot of techniques, of applications

For the (social) sciences, two main uses

- **Supervised ML: y , and a set of x**

 - ~ close to regression

 - Main output: prediction

 - Comparison with regressions ([Boelaert & Ollion, 2018](#))

WHAT IS MACHINE LEARNING?

What does machine learning do?

A lot of techniques, of applications

For the (social) sciences, two main uses

- **Supervised ML: y , and a set of x**
- **Unsupervised ML: only a set of x**
 - ~ close to GDA, clustering, etc.

Main output: clusters, reduced spaces

2 ALGORITHMS IN DETAIL

NON SUPERVISED ML

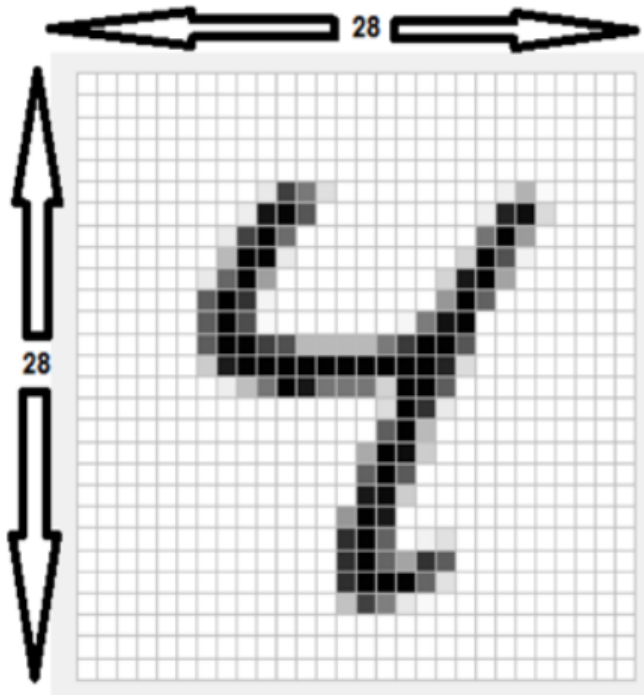
- An old endeavor, which has become popular again in the last decade
- Many techniques
- Today, 2 algorithms: Self Organizing Maps and t-SNE

ML IN A NON SUPERVISED CONTEXT

Impressive results

A classic benchmark: MNIST

Ex. 1: MNIST (10.000 x 784)



10 000 handwritten numbers

Each is transformed into one vector of 784 elements (28x28)

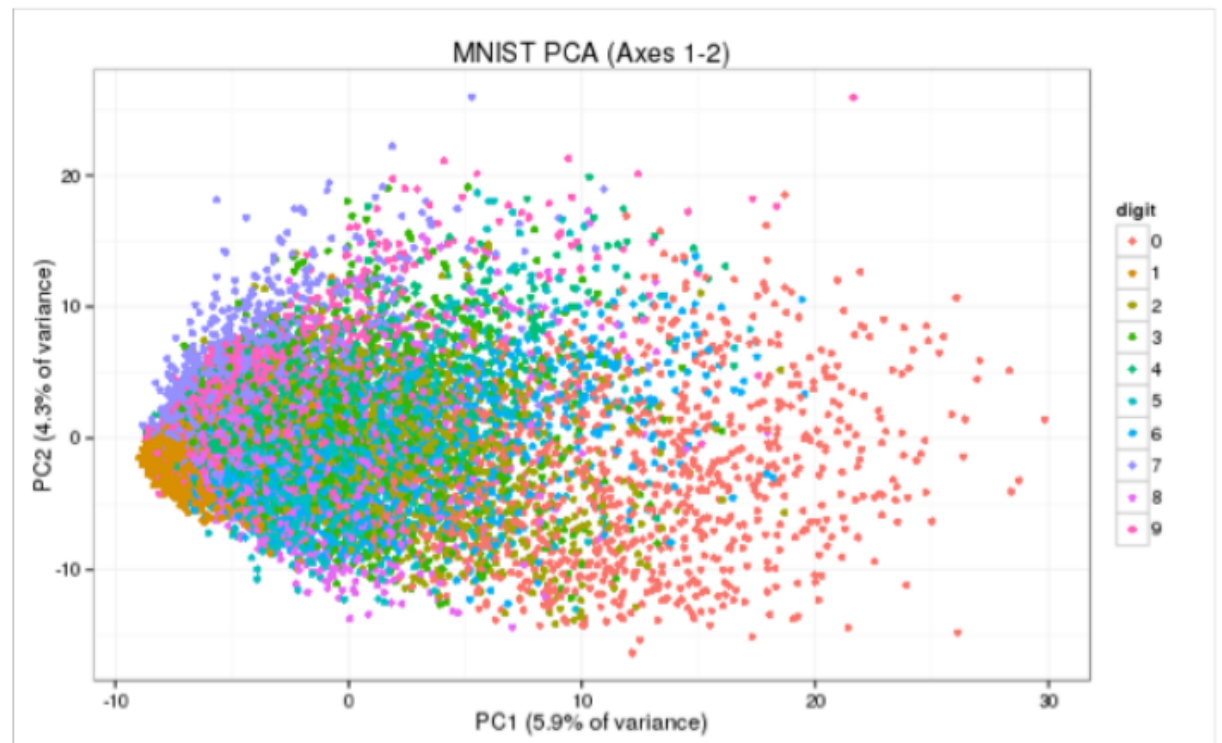
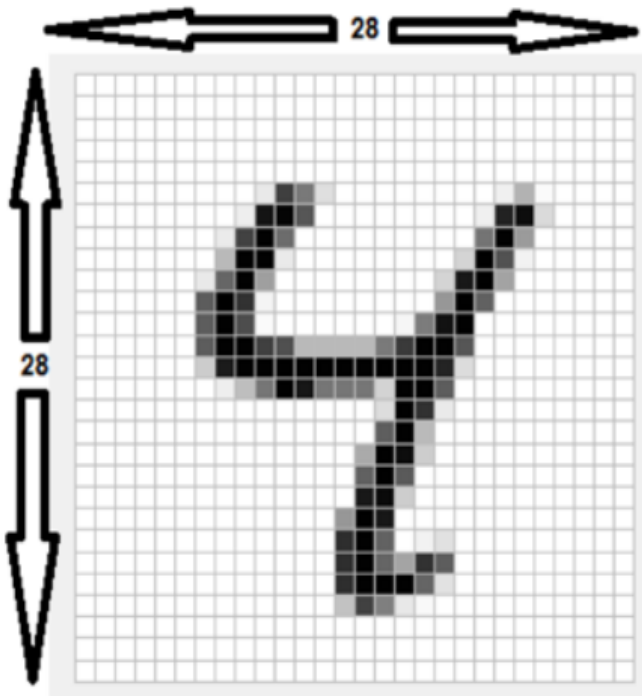
Value depends on the shade of grey

ML IN A NON SUPERVISED CONTEXT

Impressive results

A classic benchmark: MNIST

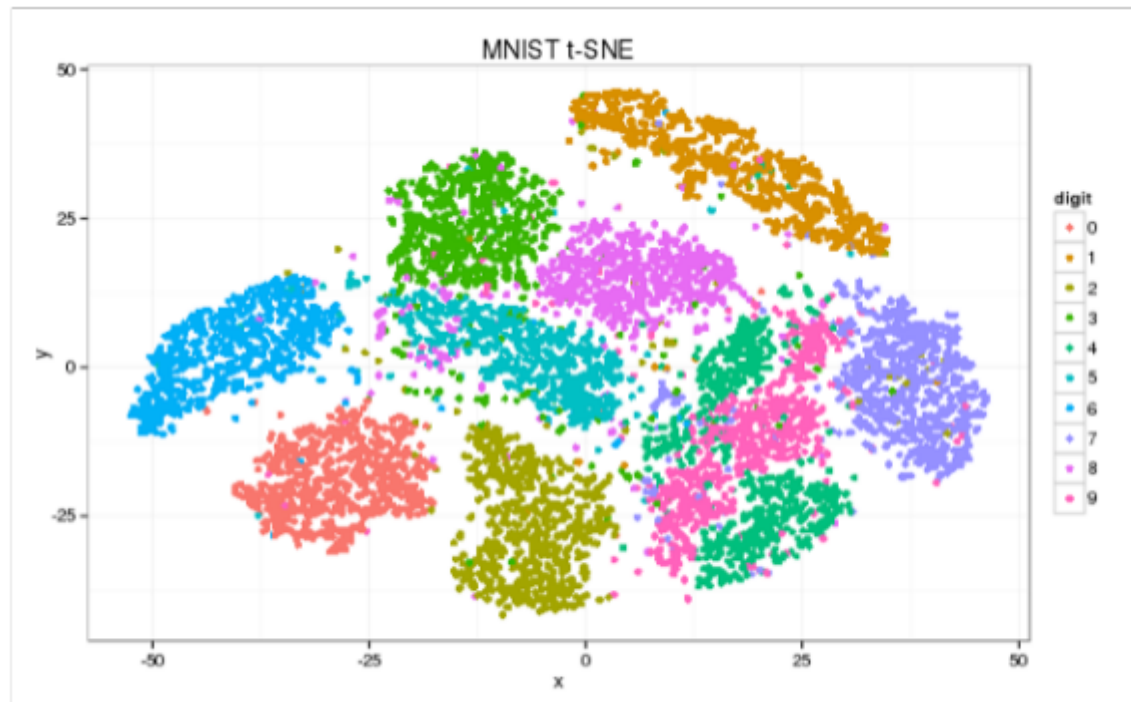
Ex. 1: MNIST (10.000 x 784)



ML IN A NON SUPERVISED CONTEXT

Impressive results

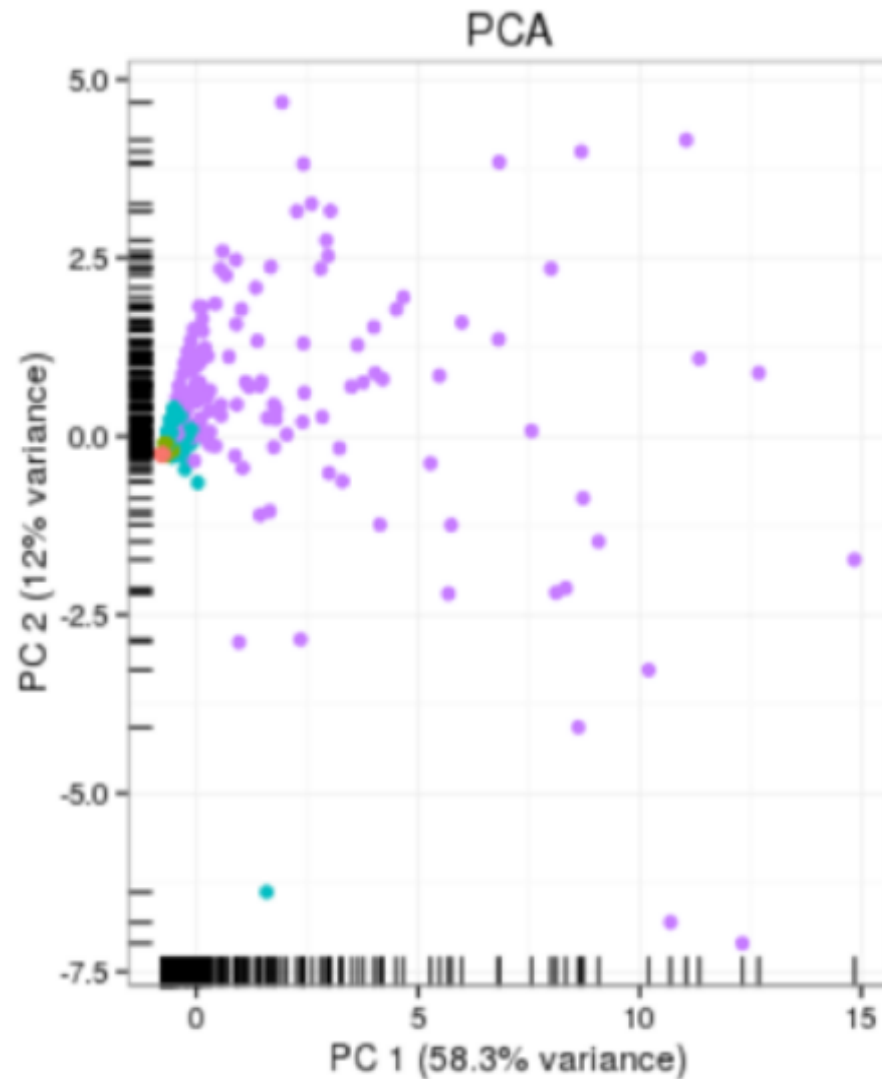
A classic benchmark: MNIST



t-SNE

ML IN A NON SUPERVISED CONTEXT

Impressive results



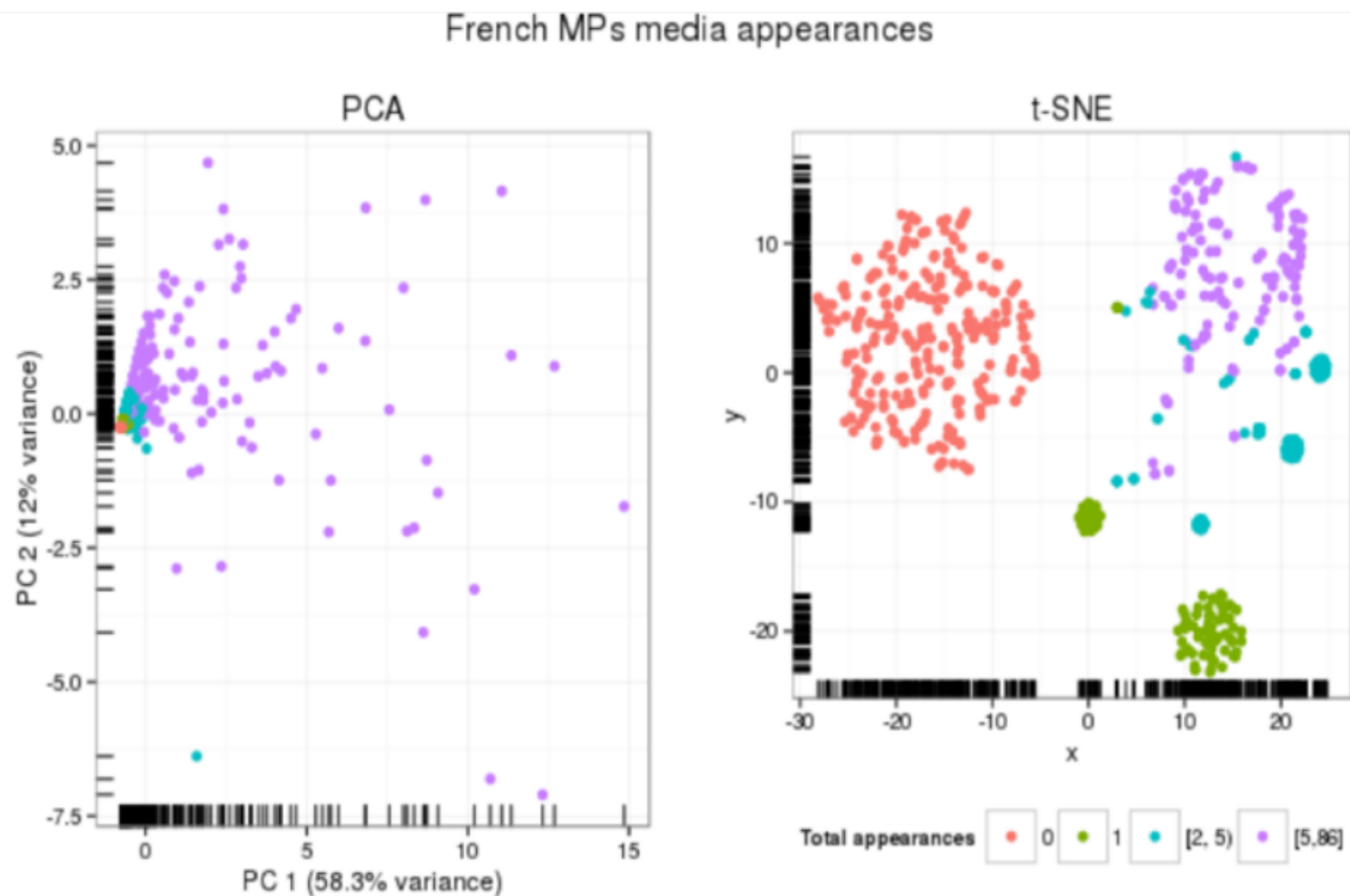
French MPs (2012-2017)

Invitation in the media

577x 7 variables (recoded)

ML IN A NON SUPERVISED CONTEXT

Impressive results



ML IN A NON SUPERVISED CONTEXT

Impressive results?

ML IN A NON SUPERVISED CONTEXT

Impressive results?

2 different types of structures captured

- The question for us: what matters most?
Local distances
Global structure?

ML IN A NON SUPERVISED CONTEXT

Impressive results?

2 different types of structures captured

- The question for us: what matters most?
Local distances
Global structure?
- Other issue: interpretation

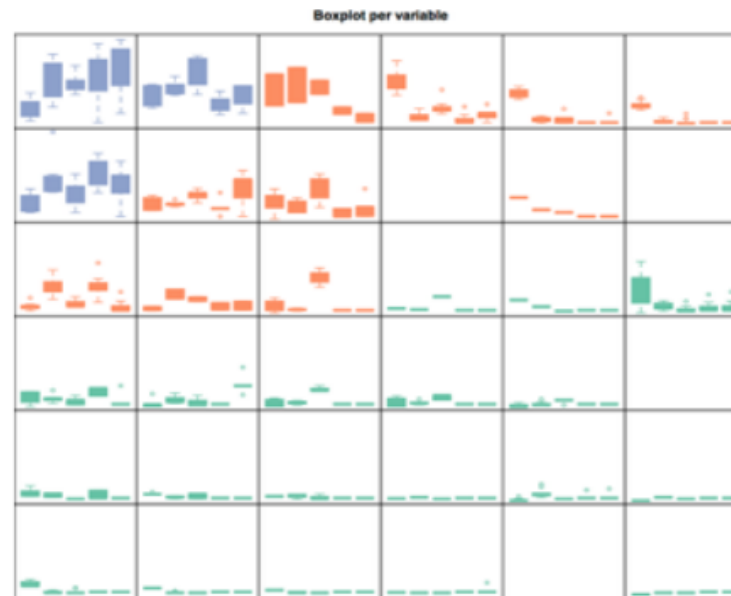
==> It is not so much « better » than just an alternative way of looking at the data set.
(Ilona's theorem)

2 ALGORITHMS IN DETAIL

SELF ORGANIZING MAPS

- Teuvo Kohonen (1982)
- 2 operations at once
 - **Dimensionality reduction**: projection of a large dimensional cloud onto a smaller plan
 - **Clustering**: Sorting the individuals into consistent groups

Generic result: a grid, with
graphs within a cell



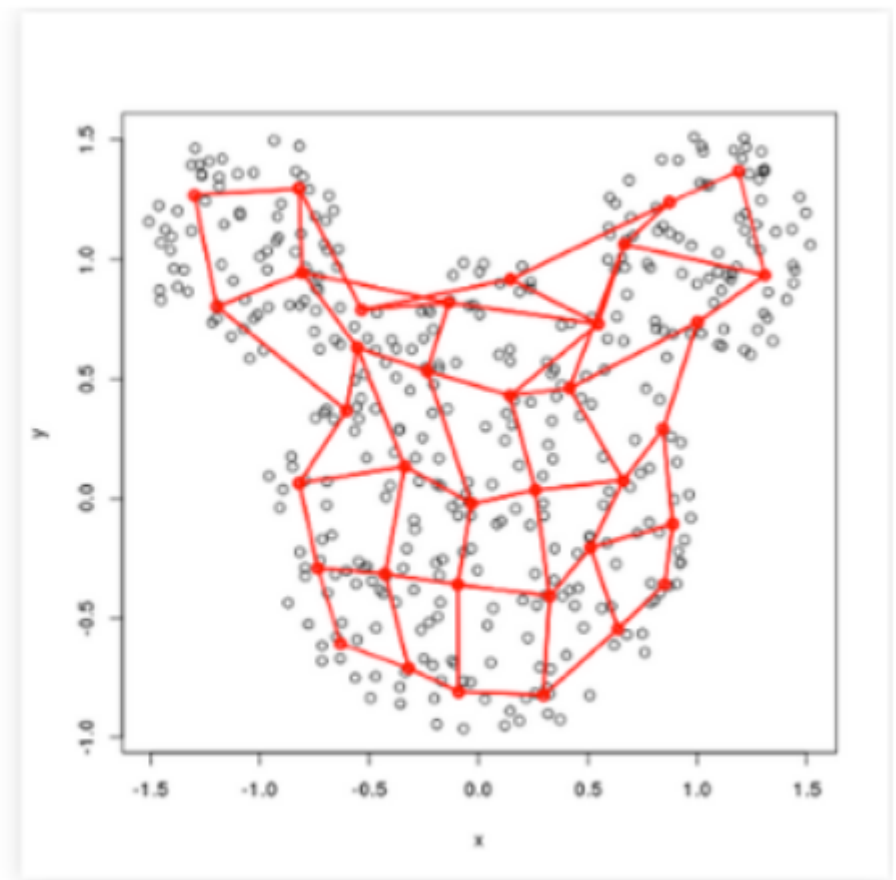
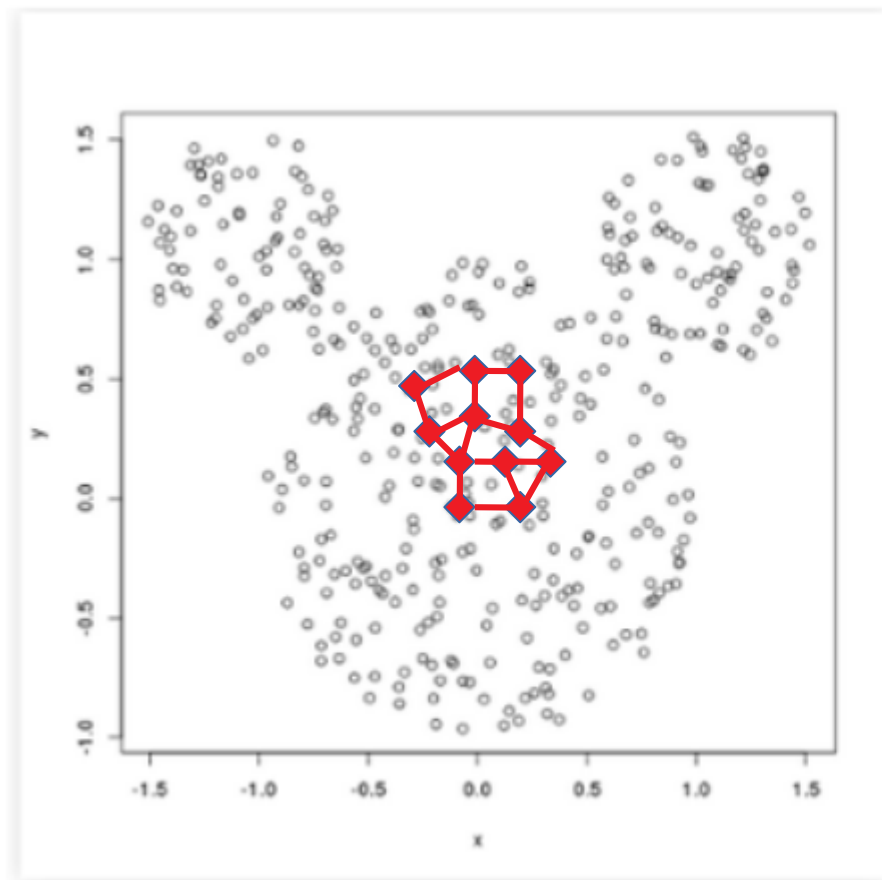
2 ALGORITHMS IN DETAIL

SELF ORGANIZING MAPS

- Principle: capture the data in the net
- Number of cells determined *ex ante*
- ...but the size of each cell is not fixed
- An important distinction: points, and prototypes

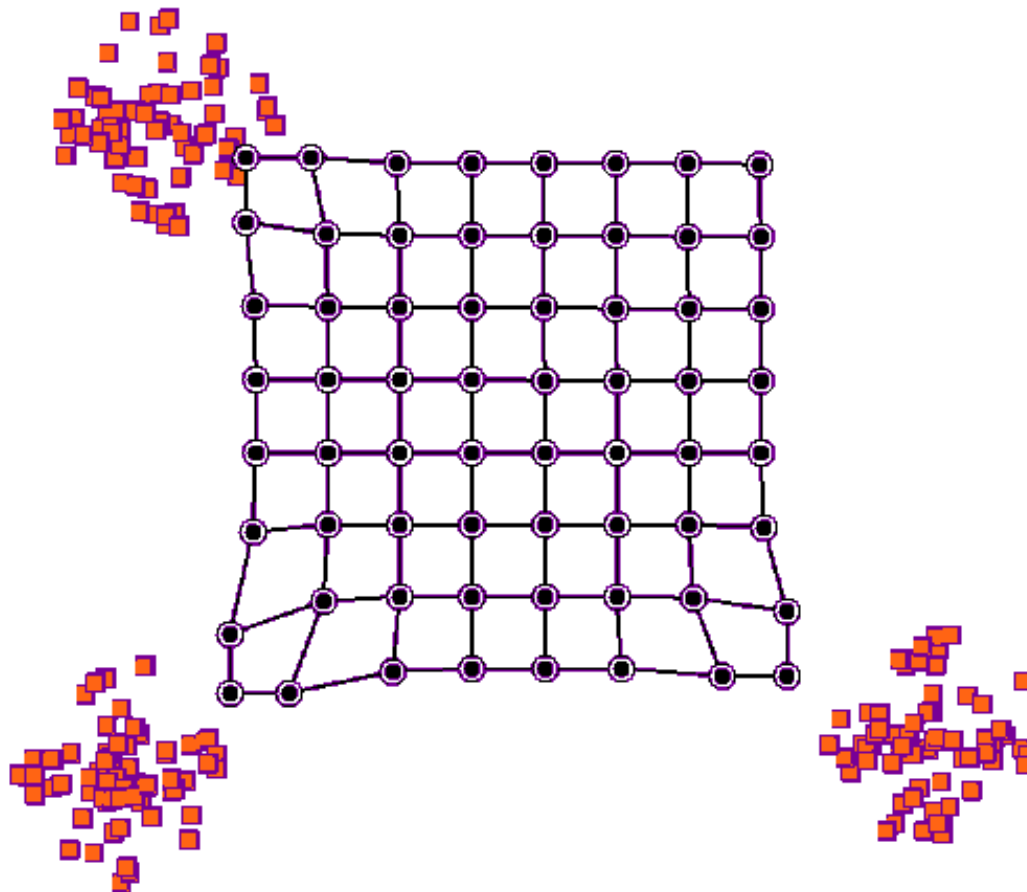
2 ALGORITHMS IN DETAIL

INTUITION



2 ALGORITHMS IN DETAIL

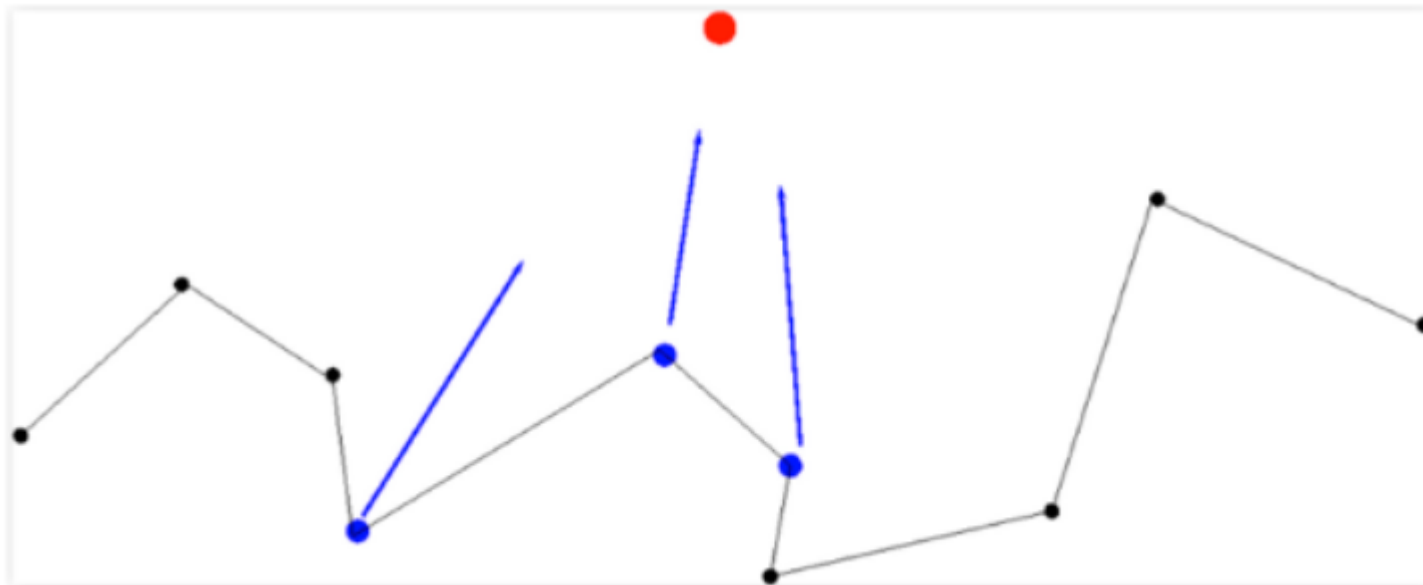
SELF ORGANIZING MAPS



2 ALGORITHMS IN DETAIL

SELF ORGANIZING MAPS

- 1) Place the prototypes in the cloud at random (or PCA)
- 2) Draw an observation $\mathbf{i}$ at random
- 3) Find the closest prototype G
- 4) Move the prototype and its $\mathbf{n}$ neighbors (net!) towards $\mathbf{i}$ (step = $\mathbf{e}$)
- 5) Repeat from 2, with $\mathbf{n}$ and $\mathbf{e}$ diminishing

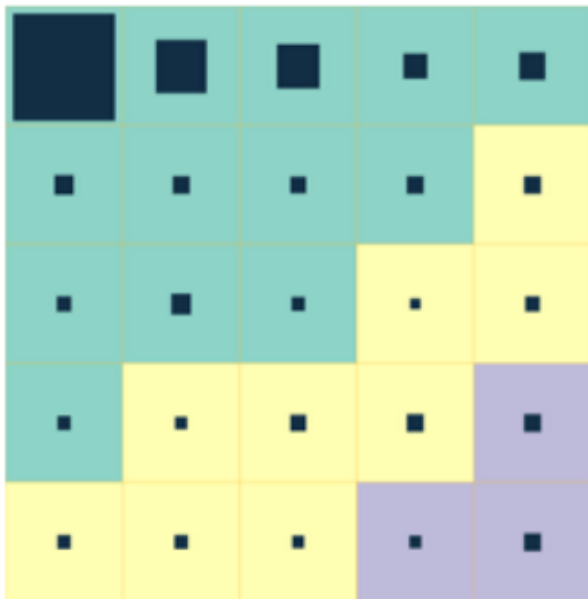


2 ALGORITHMS IN DETAIL

SELF ORGANIZING MAPS

6) When you reach stability, each point is “owned” by a prototype / is in a cell.

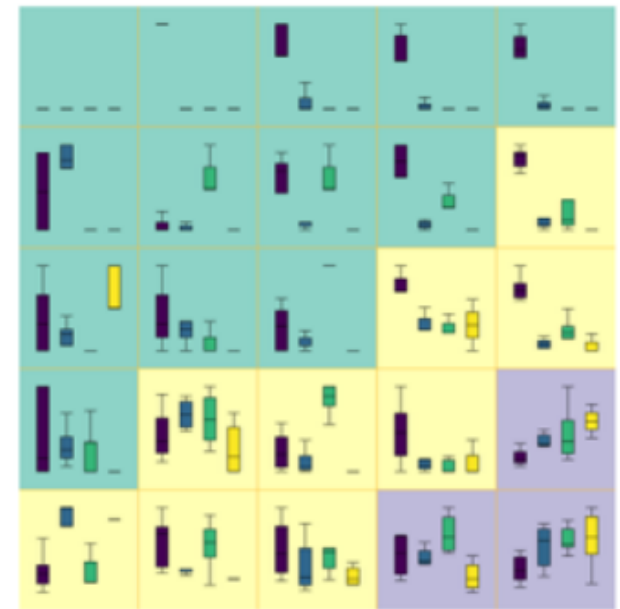
=> You can now count.



of individuals
per cell



Values of variables,
per cell

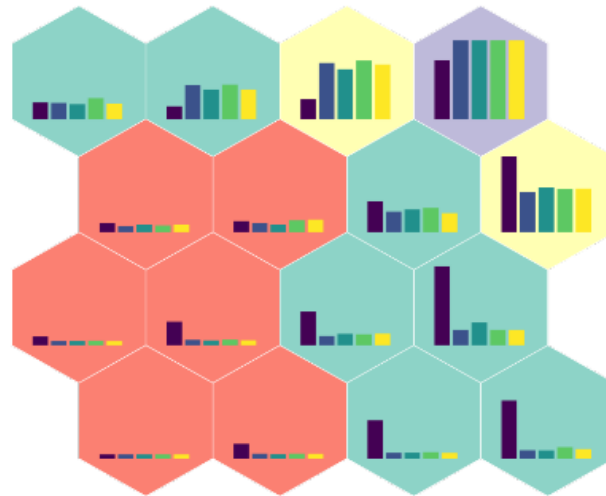


Dispersion of variables,
per cell (boxplot)

2 ALGORITHMS IN DETAIL

SELF ORGANIZING MAPS

7) You can also ask the software to do some more clustering, for instance by creating **superclasses** that divide the net in large subgroups (computed with hierarchical clustering, which measures the distance between cells).



2 ALGORITHMS IN DETAIL

SELF ORGANIZING MAPS

How to measure the quality of a projection?

Not as straightforward as with GDA, which optimizes a performance criterion (known downside of ML).

2 ALGORITHMS IN DETAIL

SELF ORGANIZING MAPS

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But

- Quantization error: Average distance between the data points and the map nodes to which they are mapped
(= small is better)

2 ALGORITHMS IN DETAIL

SELF ORGANIZING MAPS

How to measure the quality of a projection?

Not as straightforward as with GDA, which optimizes a performance criterion (known downside of ML).

But

- Topographic error: measure of how well the structure of the input space is modeled by the map (local discontinuities)
- TE is calculated by finding the best-matching and second best-matching neuron in the map for each input and then evaluating the positions. If the nodes are next to each other, then we say topology has been preserved for this input

2 ALGORITHMS IN DETAIL

SELF ORGANIZING MAPS

In R:

A couple of old packages: kohonen, SOMbrero (case sensitive!)

- A few lines of codes, a lot of options

And now: aweSOM, a graphic user interface

aweSOM: (Boelaert, Sodge, Ollion, 2020)

```
install.packages('aweSOM')
```

```
library(aweSOM)
```

2 ALGORITHMS IN DETAIL

t-SNE (t Stochastic Neighbor embedding)

- Van der Maaten & Hinton (2008)
- A star method in the early 2010s. Significant improvements on MNIST.
- « If PCA works but t-SNE does not...please check your code until you found the bug » (Van der Maaten)

2 ALGORITHMS IN DETAIL

Intuition and principles

- Idea: preserving the **local** topography
- A math-heavier technique. Transformations to achieve measures of distance, using Kullback-Leibler divergence.
- A visual technique
- Stochastic, non linear

2 ALGORITHMS IN DETAIL

t-SNE: In practice

- Works on continuous variables only
- Interpretation is possible afterwards: external variables, and groups (but pay attention: no axes, no values)
- in R: `library(Rtsne)`

Conclusion

Non supervised machine learning

- A rapidly growing area
- Carrying out classic tasks done by statistics
- Better results? It depends on your question.
- Think of combining them (denoising, etc), but under certain conditions only!